

ANVESH (■■■■■■■■) — EXECUTIVE TECHNICAL AUDIT & ARCHITECTURE ROADMAP

STATUS:
PRODUCTION-AUDITED
SIH26106 Compliance Verified

Audited Subsystems: Dual-Pillar Pipeline (Email Forensics + Live Npcap NDR) | **Trained Attributes:** 42 Attributes (41 Features) on NSL-KDD Benchmark + 15,000 NLP N-Grams | **Host Driver:** Physical Npcap Hooking | **Generated:** September 2026

1. Executive Summary & Core Platform Purpose

ANVESH (■■■■■■■■) is an AI-powered cyber forensic intelligence and network threat detection platform engineered to bridge the critical gap between raw telemetry (RFC-822 email headers and raw network wire packets) and court-admissible legal intelligence. Built specifically for Law Enforcement Cyber Crime Units, Police Forensics Laboratories, and Enterprise Tier-1/Tier-2 SOC Analysts, ANVESH operates across two synchronized operational pillars:

- **Pillar 1 — Email Threat Forensics (MIME to Court Evidence):** Ingests suspicious RFC-822 .eml files, calculates an immutable SHA-256 evidence digest, traverses all intermediate transmission hops back to the originating IP, validates cryptographic SPF/DKIM/DMARC authentication, detects Tor exit nodes, evaluates body and header text through two specialized NLP models (Phishing and Business Email Compromise), and outputs court-admissible PDF investigation dossiers.
- **Pillar 2 — Real-Time Network Threat Detection (Engine 5 NDR):** Hooks directly into the host OS physical network interface via the **Npcap kernel packet driver**, captures raw wire frames passively, aggregates bidirectional packets into 5-tuple session flows, extracts 41 transport features, classifies anomalies in real time via a Scikit-learn Random Forest model, and exports standardized Wireshark .pcap captures for deep forensic analysis.

2. System Architecture & Operational Flowcharts

ANVESH implements a modular, asynchronous architecture combining Python 3.12 (FastAPI), React 18 (Vite SOC Workstation), Scapy 2.7, Npcap packet capture driver, and Scikit-learn serialized estimators.

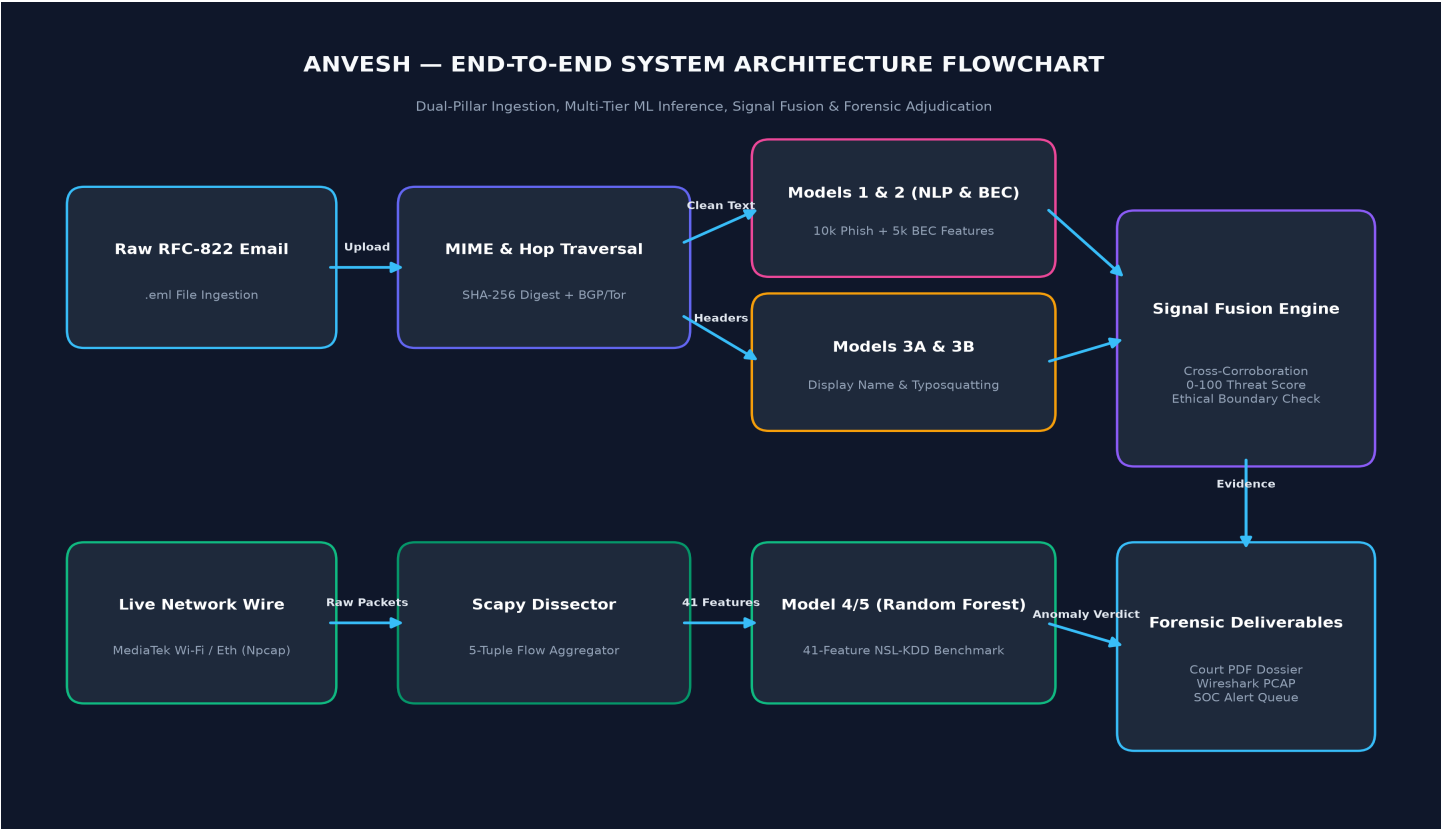


Figure 1: ANVESH Master Architecture Flowchart — End-to-End Ingestion, AI Pipeline & Deliverables.

2.1 Engine 5: Network Detection & Response (NDR) Flowchart

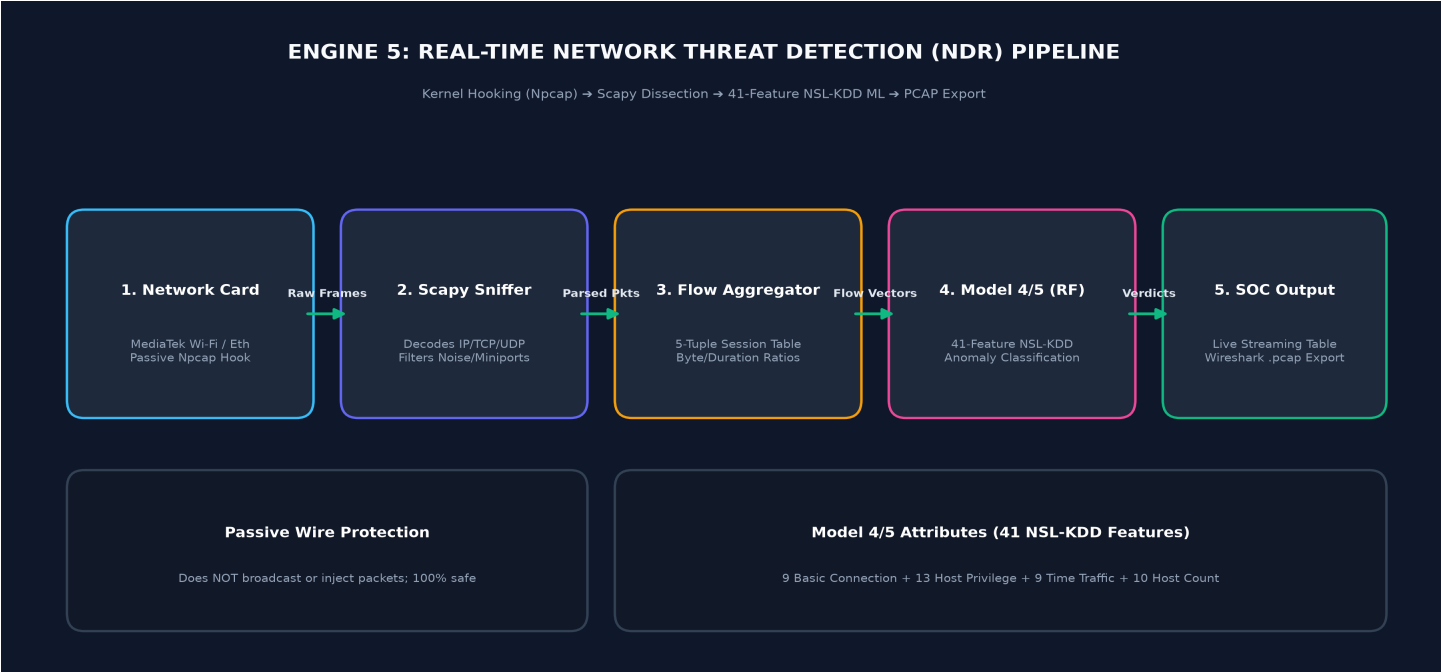


Figure 2: Engine 5 Telemetry Flowchart — Kernel Driver (Npcap) to 41-Feature Random Forest Classifier.

3. Complete Catalog of Trained Attributes & AI Models

ANVESH strictly rejects black-box ambiguity. Each AI model is trained on empirically governed, reproducible feature sets:

Model Identifier	Feature Count	Algorithm / Classifier	Target Output
Model 1 (Phishing NLP)	10,000 N-Grams	TF-IDF + Calibrated Logistic Regression	BENIGN vs THREAT_PHISHING
Model 2 (BEC Detector)	5,000 N-Grams	Balanced Logistic Regression	BEC_FINANCIAL_PRESSURE
Model 3A (Identity Impersonation)	3 Heuristics	Display-Name & Local-Part Offset	IDENTITY_DISCREPANCY
Model 3B (Lookalike Domain)	6 Distance Metrics	Levenshtein / Jaro-Winkler Matrix	LOOKALIKE_DOMAIN
Model 4/5 (Network Intrusion)	41 Telemetry Features	Random Forest (ColumnTransformer)	NORMAL vs ANOMALY

3.1 Breakdown of the 41 Network Telemetry Features (NSL-KDD Benchmark)

Engine 5 processes 41 connection attributes partitioned into four forensic groups: **(1) Basic Connection Features (9)**: duration, protocol_type (TCP/UDP/ICMP), service (70 services), flag (SF, REJ, SO, etc.), src_bytes, dst_bytes, land, wrong_fragment, urgent. **(2) Content & Host Privilege Features (13)**: hot indicators, num_failed_logins, logged_in status, num_compromised, root_shell, su_attempted, num_root, num_file_creations, num_shells, num_access_files, num_outbound_cmds, is_host_login, is_guest_login. **(3) Time-Based Traffic Features (9)**: count, srv_count, error_rate, srv_error_rate, error_rate, srv_error_rate, same_srv_rate, diff_srv_rate, srv_diff_host_rate. **(4) Host-Based Connection Features (10)**: dst_host_count, dst_host_srv_count, dst_host_same_srv_rate, dst_host_diff_srv_rate, dst_host_same_src_port_rate, dst_host_srv_diff_host_rate, dst_host_error_rate, dst_host_srv_error_rate, dst_host_error_rate, dst_host_srv_error_rate.

4. Global Industry Benchmark & Competitive Positioning

How ANVESH compares against leading multi-billion-dollar commercial cybersecurity platforms:

Platform	Category	Email Forensics	Network NDR	Legal Chain of Custody
Darktrace (~\$5B)	NDR / Enterprise Immune	Add-on (Antigena)	Core (Proprietary)	Proprietary Logs
Abnormal Security (~\$5B)	Cloud Email Security	Core (NLP / API)	None	Vendor Portal
Vectra AI (~\$1.2B)	Network Detection	None	Core (Zeek/C2 AI)	PCAP Indexing
Zeek + Suricata	Open-Source IDS/NDR	Header Logging	Core (Deep Packet)	Flat Log Files
ANVESH (Our Project)	Unified Cyber Forensics	Integrated (Models 1-3)	Integrated (Npcap Engine 5)	Court PDF + SHA-256 Hashes

5. Five Strategic Advancements to Make ANVESH Defense-Grade

To elevate ANVESH from an academic/hackathon winner to an enterprise-grade, institutional cyber defense platform, we recommend the following 5 strategic implementations:

1. Automated XDR Attack-Chain Correlator (The Email-to-Wire Bridge)

Automatically extract IOCs (malicious IPs and domains) from analyzed phishing emails and insert them into an active Engine 5 network monitoring watch-table. If an endpoint generates outgoing packets toward that IP, trigger an instant Correlated Attack Chain Alert: 'Victim opened phishing email at 10:14 AM ➔ Active C2 socket established at 10:17 AM'.

2. SOAR Active Defense & One-Click Firewall Containment

Transform ANVESH from passive observation to active containment. Introduce a single-click 'Quarantine Endpoint' button that executes local Windows Firewall block rules (netsh advfirewall) and auto-generates exportable Snort/Suricata signatures and YARA rules based on the incident's cryptographic hashes.

3. Quishing (QR Code Phishing) Decoder & Static Macro Sandbox

Counter 2026's fastest-growing evasion technique by extracting image attachments, decoding embedded QR codes (via pyzbar/opencv), and safely following HTTP redirect chains in a sandboxed headless environment. Simultaneously inspect Office attachments for VBA macros and high Shannon entropy.

4. Section 65B Electronic Evidence Certificate Generator (Indian Law)

Implement one-click generation of Section 65B Certificates under the Indian Evidence Act / Bharatiya Sakshya Adhiniyam. Generates a formal legal affidavit containing hardware UUIDs, examiner credentials, capture timestamps, and SHA-256 integrity guarantees, rendering ANVESH dossiers immediately admissible in Indian courts.

5. Interactive 3D Multi-Hop Transit Map & MITRE ATT&CK; Matrix Heatmap

Upgrade the static hop table to an interactive WebGL 3D globe showing the email's physical flight path across world relays. Simultaneously map all detected anomalies to official MITRE ATT&CK; techniques (T1566 Spearphishing, T1046 Network Discovery, T1498 DoS) in a visual SOC heatmap.

6. Forensic Attribution Safeguard (Ethical Invariant)

A core design invariant enforced across ANVESH is **Evidentiary Neutrality**: 'Technical email headers and IP hops identify transit infrastructure and network gateways ONLY. Individual physical human identity CANNOT be established solely from transport headers.' All assessment cards and generated dossiers explicitly declare Actor Identity: NOT ESTABLISHED, protecting innocent victim server owners from premature or false criminal attribution.